Iteration 2 Report

Meal Planner App for the Elderly (MealCare)

Programme: B.Sc. in Software Development

Module: Third Year Software Development Project

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Submission Date: 25 March 2026

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1. Iteration Objectives

Iteration 2 covers development from after 3 March 2026 through the cur- rent milestone. The main goals were to move the app from a locally stored prototype to a persistent, user-specific system: secure sign-in and sign-up, profiles and meal plans stored in Supabase, integration with a live recipe API (Spoonacular) aligned with user preferences, a View Plan experience for the saved week, meal completion tracking, and shopping list generation with cloud sync, filtering by aisle, and check-off behavior. Secondary goals included keeping the layered architecture (context, services, hooks, pages) so Iteration 3 can add polish and any remaining features .

2. Features Implemented

2.1. Authentication and Cloud Persistence (Supabase)

Email and password sign-in and sign-up are implemented with Supabase Auth, including session loading on app start and an onAuthStateChange listener so the UI stays in sync when the user logs in or out. Profiles are no longer only local: AppContext loads and saves profile fields (age group, dietary preferences, allergies, health conditions, budget) through profile service calls tied to the authenticated user's ID. Weekly meal plans are persisted per user and week via the meal plan service, with loading and saving coordinated through AppContext so the planner and View Plan screens share one source of truth.

2.2. Live Meal Data and Nutrition Pipeline

The app integrates Spoonacular (or equivalent API flow in the codebase) so meals can be fetched using profile-aware query parameters (diet, intolerances, health-related constraints). Work in this iteration included normalizing nutrition-related data, tightening health-condition and diet-based filtering, and a nutrition evaluation layer that maps user choices to compatibility rules. API key handling was improved during development (including consideration of server-side or environment-based configuration so keys are not exposed unnecessarily).

2.3. View Saved Plan and Progress Tracking

A View Plan page loads the saved weekly plan from the database for the current week and presents it with a clear day-by-day layout. Meal completion tracking is separate from planning: users can mark individual meals (or whole days) as done, with progress and day-tab indicators reflecting completion state. Completion data is stored through a dedicated meal completion service, keeping tracking logic out of presentational components.

2.4. Smart Shopping List

The Shopping page builds a list from ingredients implied by the confirmed weekly plan, with logic to consolidate items where appropriate and persist rows in Supabase against the user's meal plan / shopping list record. If the database list is empty but the in-memory plan has meals, the app can sync items from the plan. The UI supports aisle-oriented filtering, status filtering, check-off for bought items, shopping tips, and progress feedback. The data model includes hooks for future barcode-linked items (added_by_barcode), but full camera scanning is not the focus of this iteration.

2.5. Application Architecture Updates

The architecture from Iteration 1 was extended, not replaced:

- Auth layer (AuthContext + Supabase client) centralizes session and auth actions.
- App layer (AppContext) coordinates profile hydration, meal plan ID, weekly plan state, and loaders after authentication.
- Service layer expands with profileService, mealPlanService, shoppingListService, mealCompletionService, and API-facing mod- ules.
- Hooks (e.g. meal planning and tracking hooks) keep pages thin and testable.
- This keeps concerns separated and matches the Iteration 1 intent to scale with new features.

2.6. Interface and Navigation

A dedicated login / registration experience (LoginForm) was added with themed styling consistent with an elderly-friendly, clear layout. The navbar routes users to Home, planning, browsing, View Plan, Shopping, Profile, and Login. The meal plan screen received responsive layout improvements and safer optional handling so the UI does not crash before selections exist. CSS variables and mobile rules reduce hard-coded colors and improve maintain- ability.

3. React and JavaScript Concepts Used

Throughout this iteration, the following were applied:

- React Hooks: useState, useEffect, useContext, useCallback, and custom hooks for meal planning and tracking.

- Context: Splitting authentication (AuthContext) from application data (AppContext) to avoid a single overloaded provider.
- Asynchronous flows: async/await for Supabase and API calls, with cancellation flags or dependency guards in useEffect where needed to avoid stale updates.
- JavaScript collections: .filter(), .map(), merging and deduplicating ingredient-like structures for shopping lists.
- Routing: react-router-dom for protected-by-flow navigation (auth gat- ing implied by data loading in context).

4. Changes to Vision Document

The overall vision for MealCare (supporting elderly users with accessible, preference-aware meal planning) is unchanged. Emphasis in implementation shifted toward cloud-backed, per-user data and end-to-end workflows (plan → view → shop → track), which were always part of the long-term intent but were deferred in Iteration 1.

5. Changes in Research and Design (R&D)

Decisions during Iteration 2 included:

- Adopting Supabase for authentication and relational storage of profiles, meal plans, shopping lists, and completions.
- Integrating an external recipe API for richer meal data while preserving the existing filtering and nutrition rules.
- Splitting Auth from App context for clearer lifecycle: login state vs. profile and plan hydration.
- Shopping list schema prepared for future barcode use without blocking current delivery.

6. Changes in Specifications

Specifications were refined to match implemented behaviour:

- User accounts and server-side profile rows replace “local only” profile storage for logged-in users.
- Meal plan save/load is defined per week (week start alignment) and tied to mealPlanId.
- View Plan explicitly includes tracking and read-oriented display of saved days.
- Shopping list specifies generation from the weekly plan, sync when the DB is empty but the plan has meals, checked state, and aisle categorisation.
- Validation and loading states (e.g. waiting for meal plan ID before showing errors) clarify UX during async loads.

7. What Was Not Completed

Some items remain for Iteration 3 or later:

- Full barcode scanning for shopping verification (fields/UI hints exist; end-to-end scanning workflow is not delivered).
- Further accessibility audits (screen reader pass, contrast, font scaling) beyond current responsive styling.
- Optional enhancements such as push reminders, offline mode, or multi- week history views, if prioritized by the final specification.

8. Iteration Summary

Iteration 2 delivers a substantially complete vertical slice: users can register or sign in, maintain a cloud profile, build a week plan with API- backed meals and nutrition-aware filtering, view that plan, track what they have eaten, and maintain a synchronized shopping list with practical supermarket-style filters. The codebase remains layered and service- oriented, which supports the remaining scope in Iteration 3.

9. Revised Iteration 3 Plan

- Barcode-assisted shopping (if feasible): implement capture or lookup flow and connect to added_by_barcode and list update.

- Hardening: broader testing (including service unit tests where useful), error messages for network failures, and loading skeletons for slower connections.
- Accessibility and UX polish: focus on older adults—larger touch targets, contrast, simple language, and predictable navigation.
- Demonstration readiness: scripted demo data, cleanup of debug logging where appropriate, and documentation for assessors (setup of .env / Supabase project).

10. Implemented Figures

The following figures present screenshots of the implemented application at the end of Iteration 2. These images demonstrate the core functional features delivered in this iteration and confirm that a potentially shippable product has been deployed.

The **Fig.1** screen represents the main entry point of the application. It provides navigation to core features such as Profile Setup, Meal Planning, and Shopping (feature placeholders), demonstrating overall system structure and accessibility-focused layout.

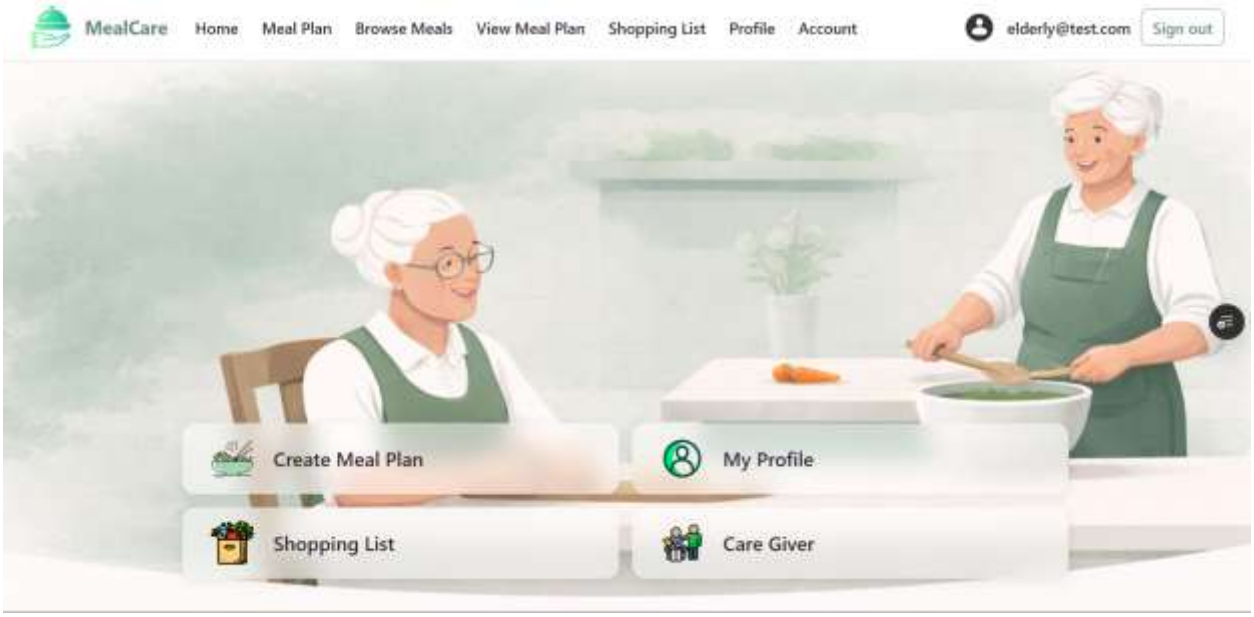

Fig.1: Home / Dashboard Screen (Implemented Version)

The **Fig.2** User profile screen showing age group, dietary preferences, allergies, health conditions, and budget. The form uses controlled inputs; data is saved to the signed-in user's profile in Supabase.

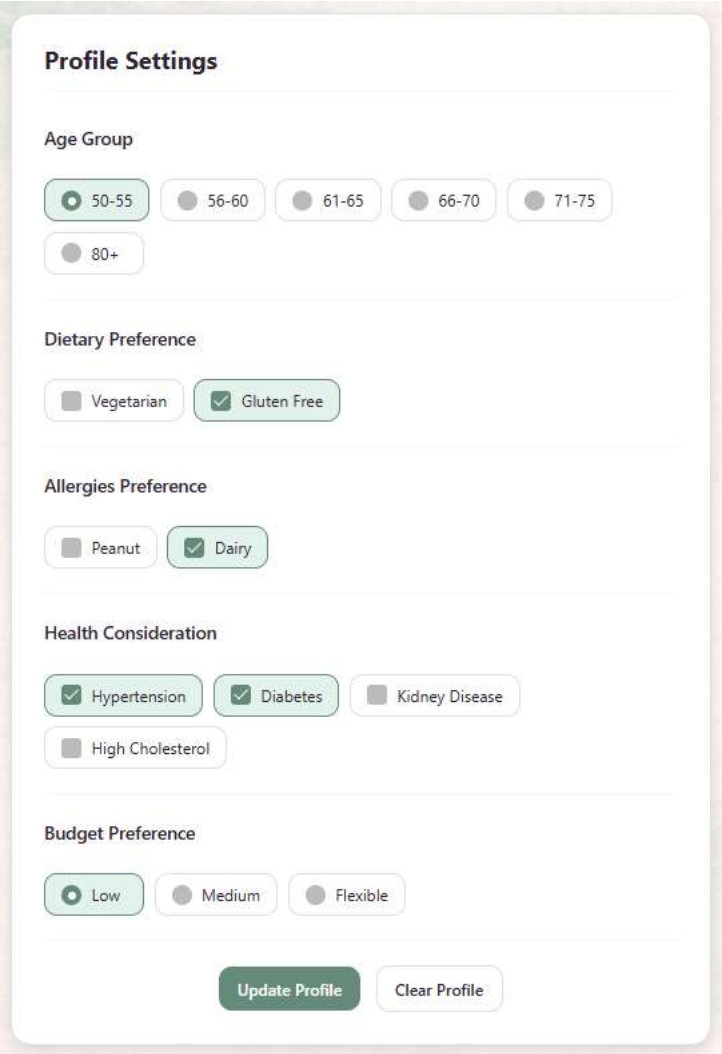

Fig.2 User Profile Management Screen (Implemented Version)

The **Fig.3** Weekly meal planning screen for Monday–Sunday, with breakfast, lunch, and dinner slots. Filtered meals from the API are shown by category; the user selects each slot and saves the day or week. A progress indicator shows how much of the week is planned.

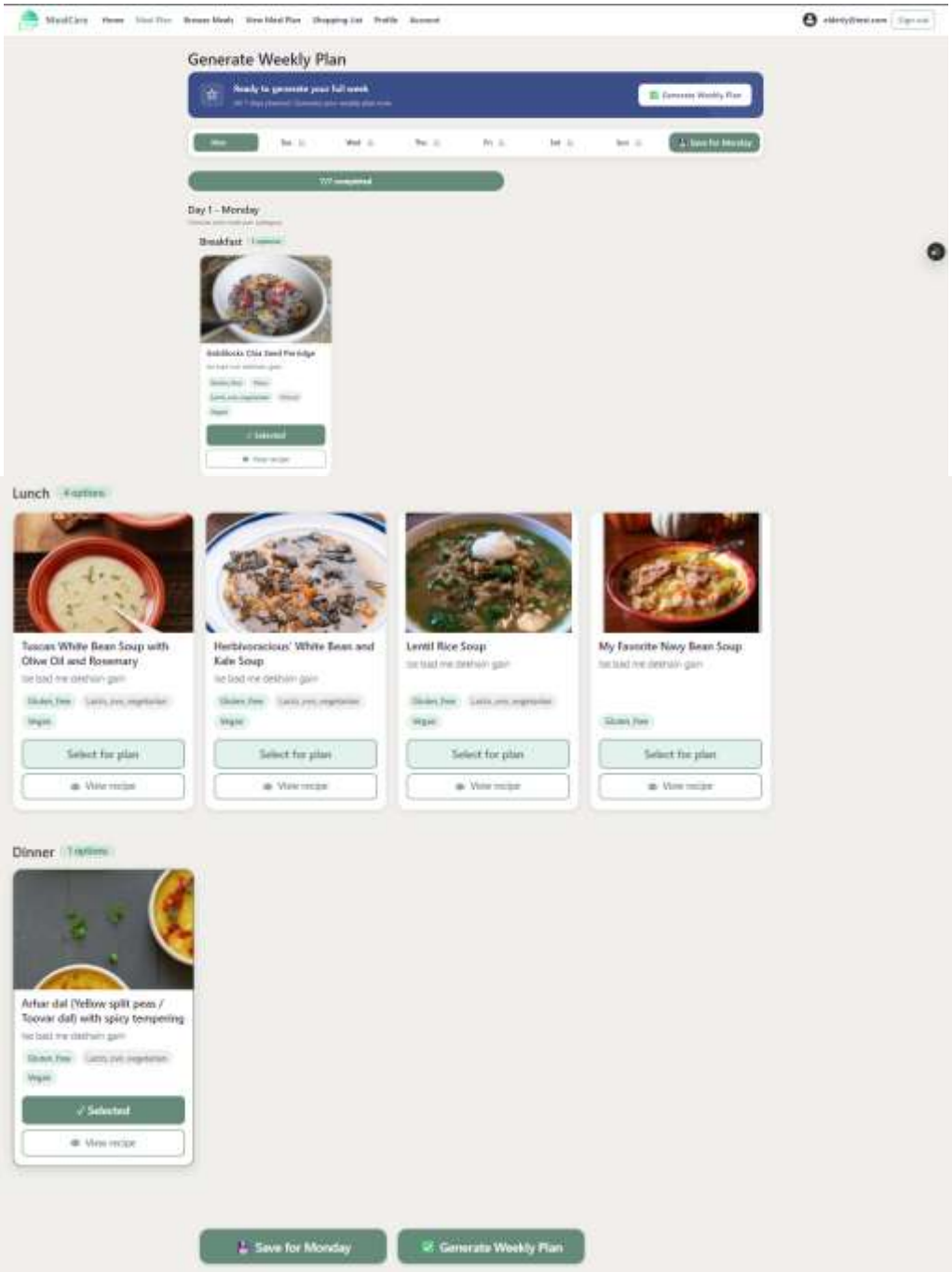

Fig.3 Weekly Meal Planning Screen (Implemented Version)

The **Fig.4 S**hopping list screen showing ingredients derived from the saved weekly plan, grouped or filtered by aisle, with checkboxes for purchased items and optional progress or tips for the shopping task.

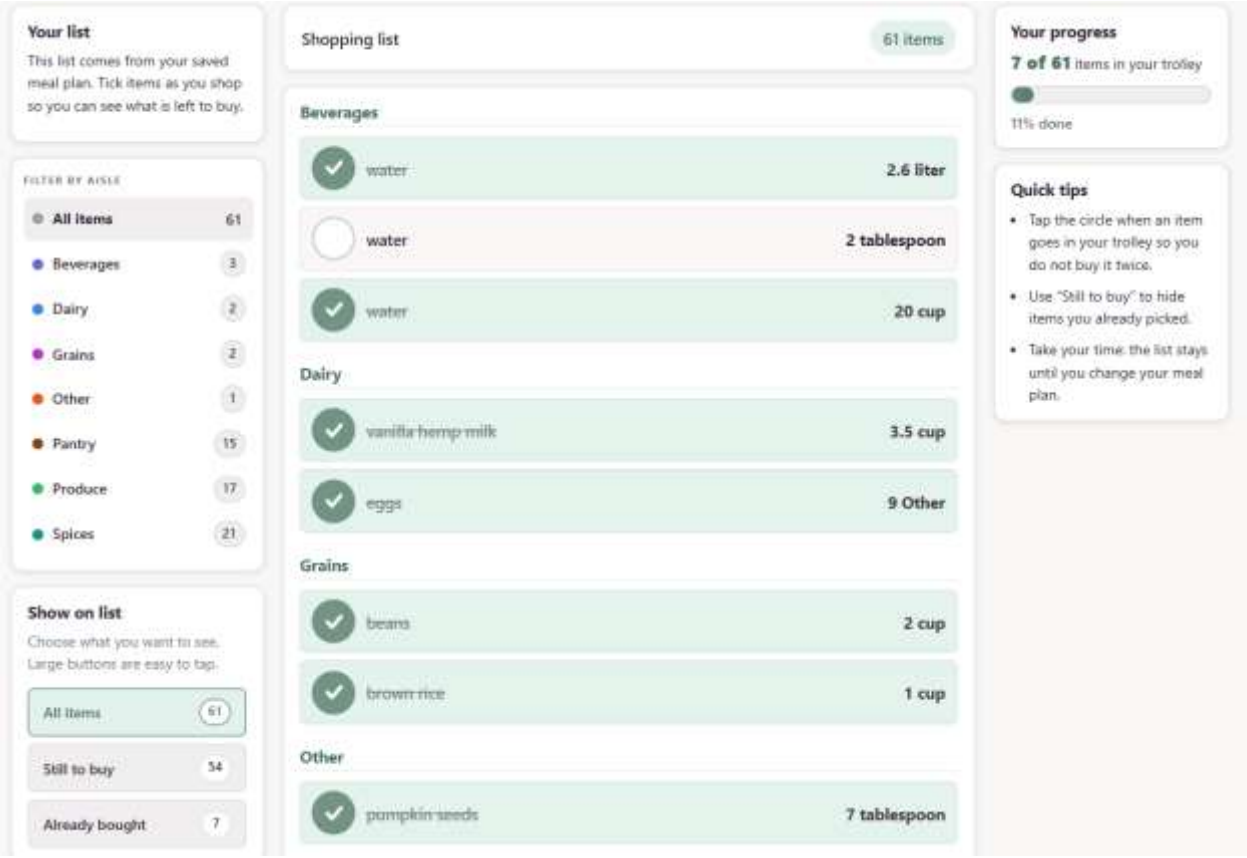

Fig.4 Shopping List Screen (Implemented Version)

The **Fig.5** View plan screen showing the saved weekly meal plan for the current week, with days selectable to inspect breakfast, lunch, and dinner. Meal completion tracking (per meal or day) and progress are visible so users can see what is planned versus already eaten.

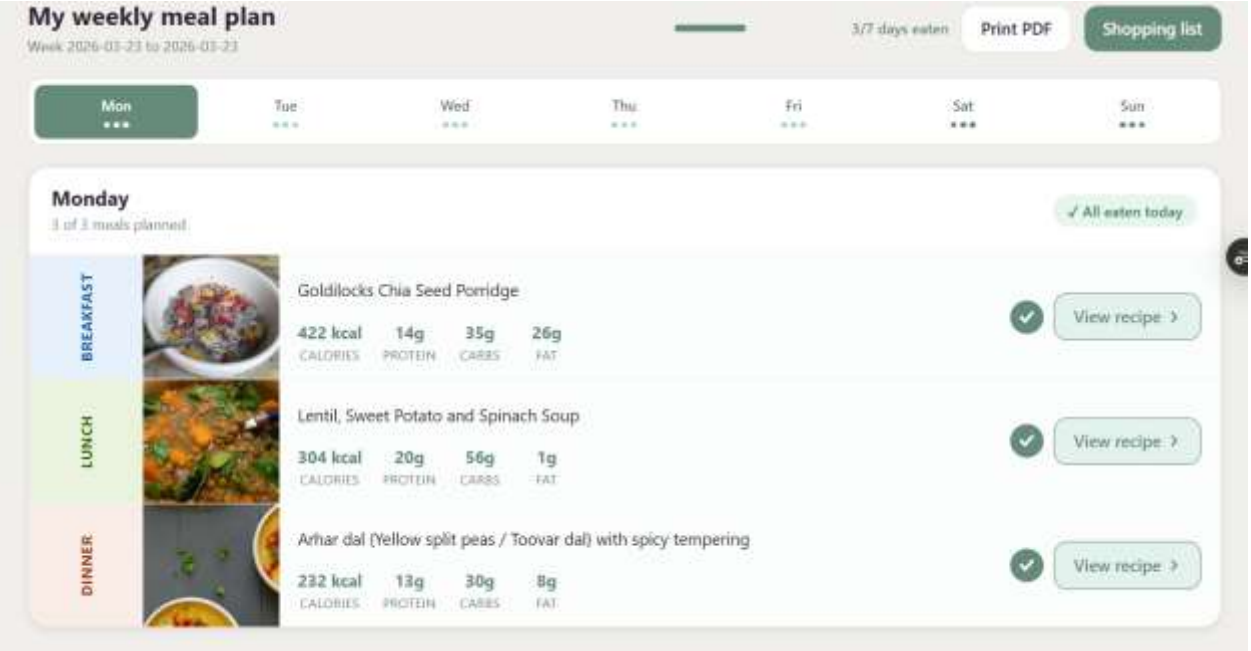

Fig.5: View Meal Plan Screen (Implemented Version)

11. Use of Generative AI

Generative AI tools were used only to assist with content structuring, idea clarification, and summarizing technical details not for direct code generation. Gen AI refers to systems that can produce text, images, or other forms of content based on learned data patterns. In this project, it supported clarity and helped with documentation preparation and code cleaning.

DECLARATION

*I declare that all material in this submission, e.g. thesis/essay/project/assignment, is entirely my/our own work except where duly acknowledged.

*I have cited the sources of all quotations, paraphrases, summaries of information, tables, diagrams or other material, including software and other electronic media in which intellectual property rights may reside.

***I have cited clearly, exhaustively and adequately in my source code, code generated by Gen AI tools.**

*I understand that I can only get credit for work that I have done (e.g. understanding other code, interfacing my code with it, modifying it, etc.), but you will not be credited with the third party code itself.

*I have provided a complete bibliography of all works and sources used in the preparation of this submission.

*I understand that failure to comply with the Institute's regulations governing plagiarism constitutes a serious offence.

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